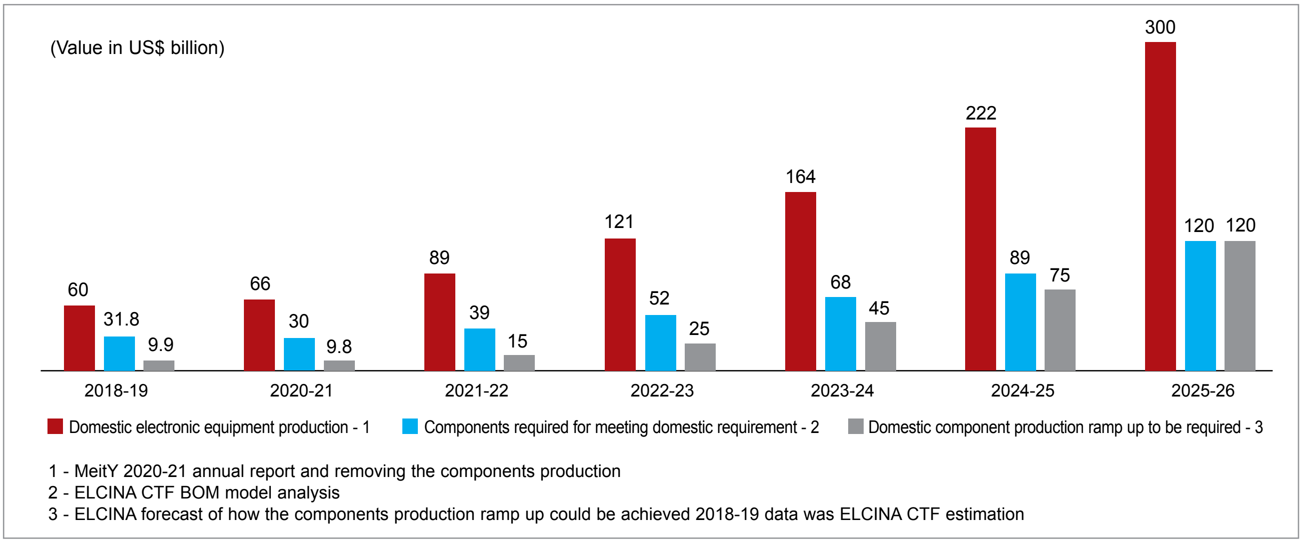

Fig. 2: Potential for manufacturing electronic components in India

2022-23. To meet this demand, $52 billion worth of components were required but only $3 billion worth of components were produced domestically.

Keskar explains, "In any electronics product, components contribute approximately 40% of the total value, 25% for semiconductors, 10% for passives, connectors, and cables, and 5% for bare PCBs. With a projected $300 billion growth potential, the demand for passive components and bare PCBs is expected to reach around $30 billion and $15 billion, respectively, both of which are predominantly imported currently. Unlike semiconductor fabrication labs, assembly, testing, marking, and packaging (ATMP) facilities, passive components, and PCBs do not require massive investments."

Global electronics products rely heavily on passive components, accounting for about 48% of the components, and active components comprise the remaining 52%. The Confederation of Indian MSMEs in ESDM and IT (CIMEI) Director-General Jairaj Srinivas believes India can manufacture simple components such as transformers, capacitors, and transistors. "These components do not require extensive expertise. By encouraging and supporting local production of these passive components, India can create a self-sustaining ecosystem for the electronics industry and reduce dependence on imports, fostering domestic manufacturing," he adds.

Currently, the government has introduced the Scheme for Promotion of Manufacturing of Electronic Components and Semiconductors (SPECS) to support the manufacture of passive components and the PCB sector. "Under this scheme, a 25% incentive on capital expenditure investment is offered for manufacturing these products, following a reimbursement policy. However, there is a need to modify this scheme to encourage wider participation, including MSME industries. With a projected $45 billion opportunity in the next five years, this sector holds immense potential and can significantly contribute to India's electronics manufacturing capabilities," Keskar adds.

Salvaging the squirming sector

The domestic electronics manufacturing sector is squirming to compete with the likes of Wistron and Foxconn, who, with their gargantuan resources, have set up massive assembly units but continue to import essential electronics components. Creating a level playing field for domestic manufacturers by reducing the bureaucratic red tape for infrastructure expansion, boosting intellectual property (IP) creation and investment in research and development of embedded electronics products can contribute to almost 30% of the value in the electronics industry.

Infrastructure. These opportunities can be availed to ensure the success of Indian MSMEs that design electronics products in India and electronics design-as-a-service startups, by creating pathways to quality infrastructure and access to funding to establish huge manufacturing facilities to compete with the industry behemoths.

In April 2020, the government of India announced the Electronics Manufacturing Cluster 2.0 (EMC 2.0) Scheme to establish a complete ecosystem for ESDM manufacturing/production. While many may welcome this move, the EMC 2.0 Scheme fails to provide specific details about whether it provides investment for